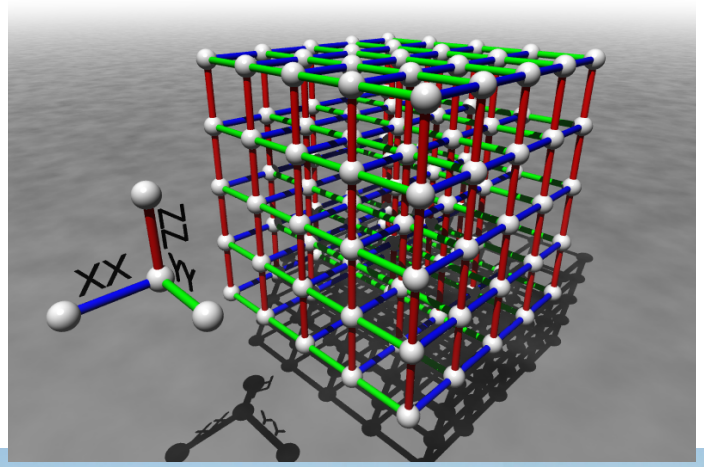
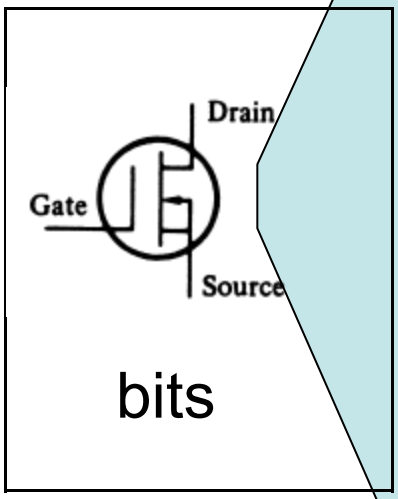
How to Build a Quantum Computer

Dave Bacon
Department of Computer Science & Engineering
University of Washington, Seattle, WA

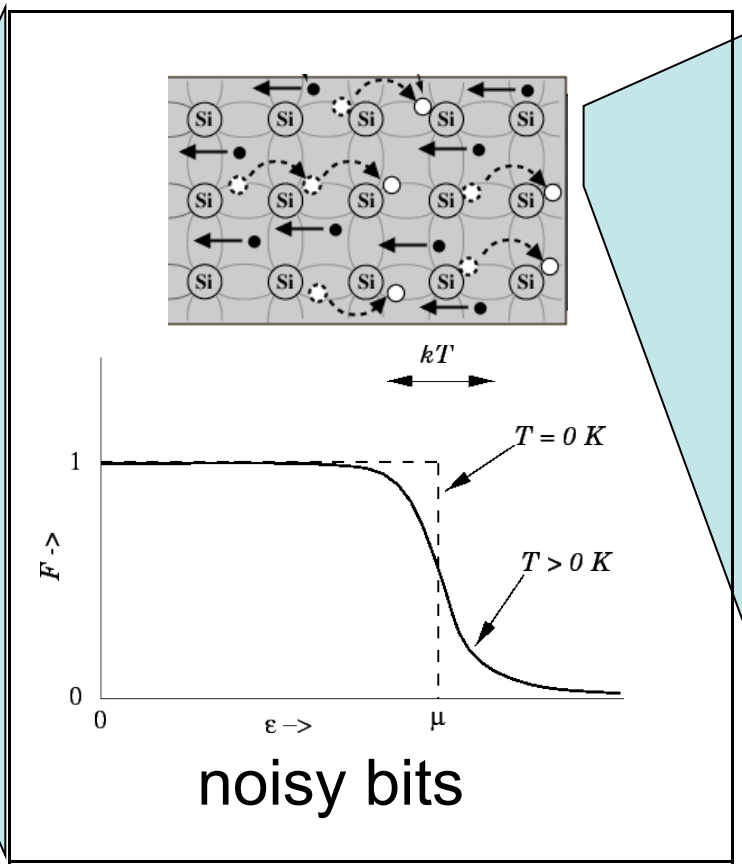




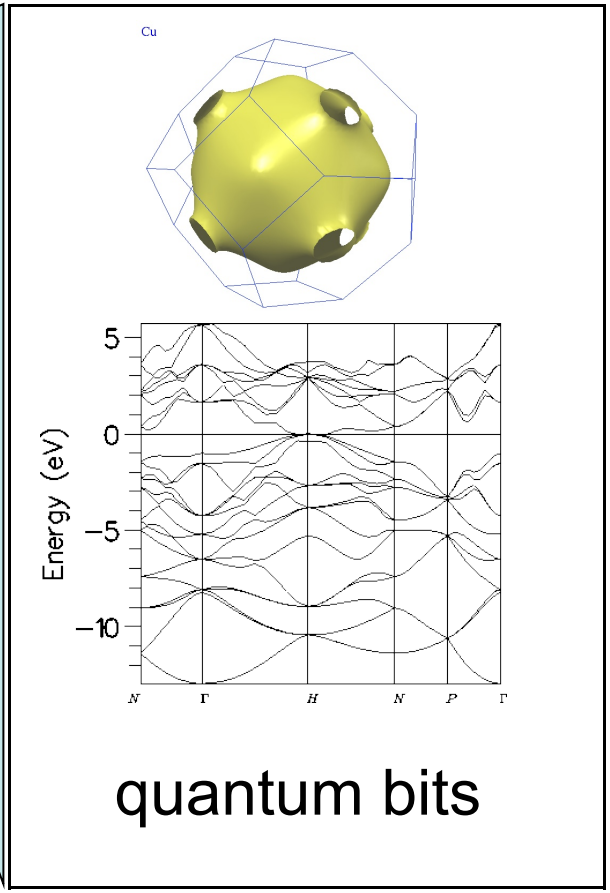
Behind the Bit



bits



noisy bits



quantum bits

deterministic

probabilistic

quantum

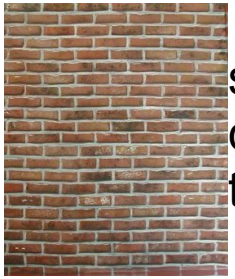
Quantum Computer?



Richard Feynman

"How can we simulate quantum mechanics? ...Can we do it with a new kind of computer… a quantum computer? It's not a Turing machine but a machine of a different kind" - Richard Feynman (1981)





simulating quantum theory

0	1	0
1	1	0
0	1	1

deterministic bit configuration

?	?	?
?	?	?
?	?	?

probability of bit configuration

"real positive number"

?	?	?
?	?	?
?	?	?

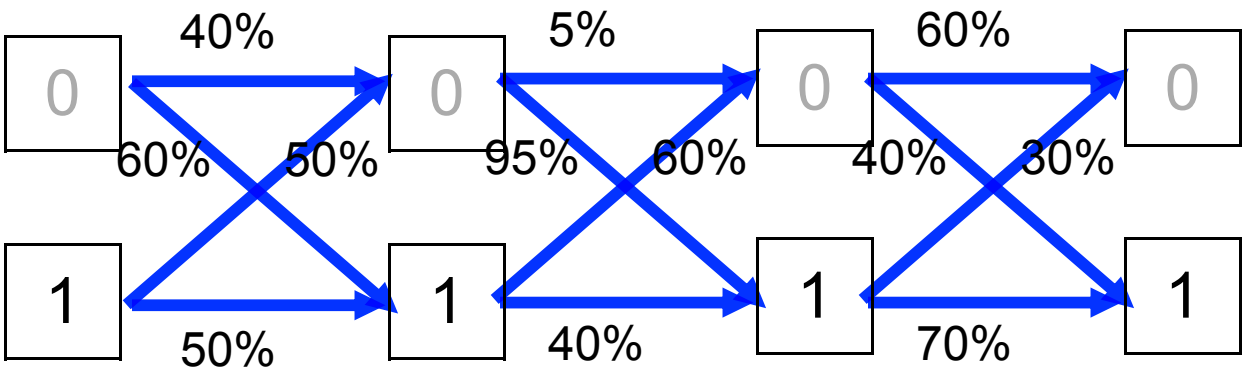
amplitude of bit configuration

"real number"

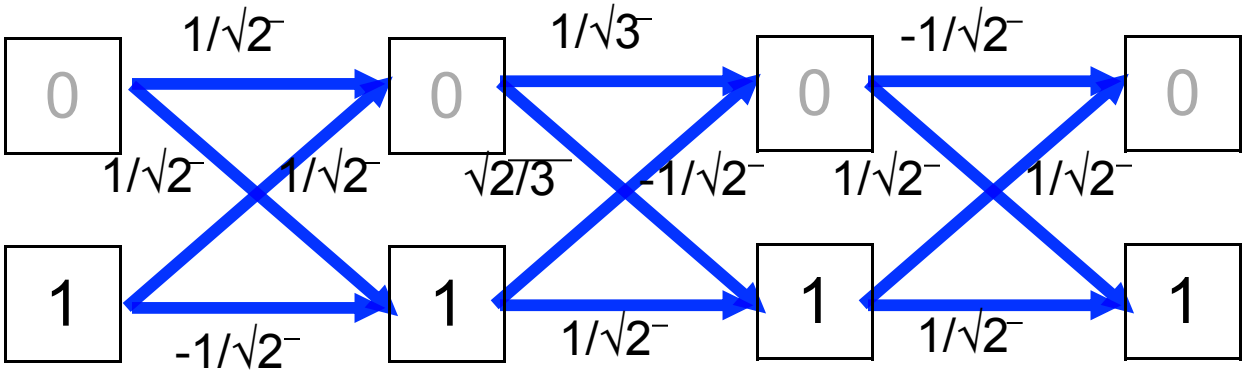
Probability ="square of real number"

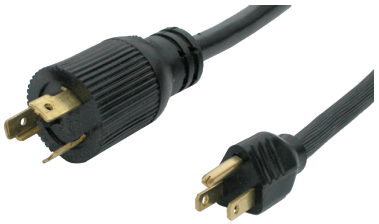
Subtracting Computers

Probabilistic



Quantum





What Power These Negative Numbers?

Quantum Computers!

188198812920607963838697239461650439807163563379417382700763356422988859715234654853190606065047430453173880113033967161969232120573403187955065699622130516875930765025705 9 = 398075086424064937397125500550386491199064362342526708406385189575946388957261768583317 × 472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

	Classical	Quantum
FACTORING	$O(\exp(n^{1/3}))$	$O(n^2 \log n)$



Public Key Cryptosystems

RSA
Diffie-Hellman
Elliptic Curves

BROKEN



Quantum Computers!

Quantum algorithms that offer speedups over classical algorithms

- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]
- Pell's equation [Hallgren 02]
- Hidden shift problems [van Dam, Hallgren, Ip 03]
- Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]
- Simulation of quantum systems



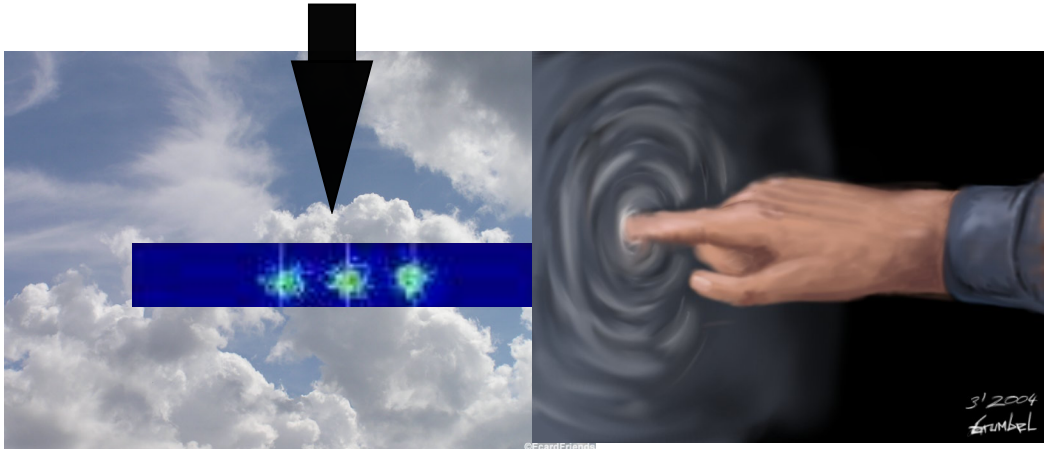
Applications to chemistry,
biology, material science….

Why Aren't All Computers Quantum?



No Information without Physical Representation!

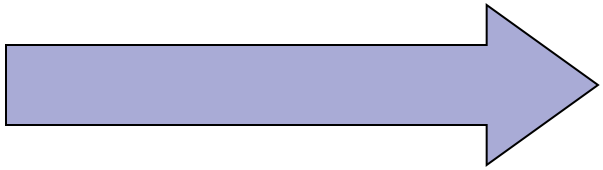
Quantum information



environment control

?	?	?
?	?	?
?	?	?

quantum

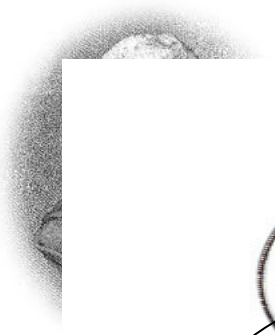


?	?	?
?	?	?
?	?	?

classical

Why Aren't All Computers Quantum?

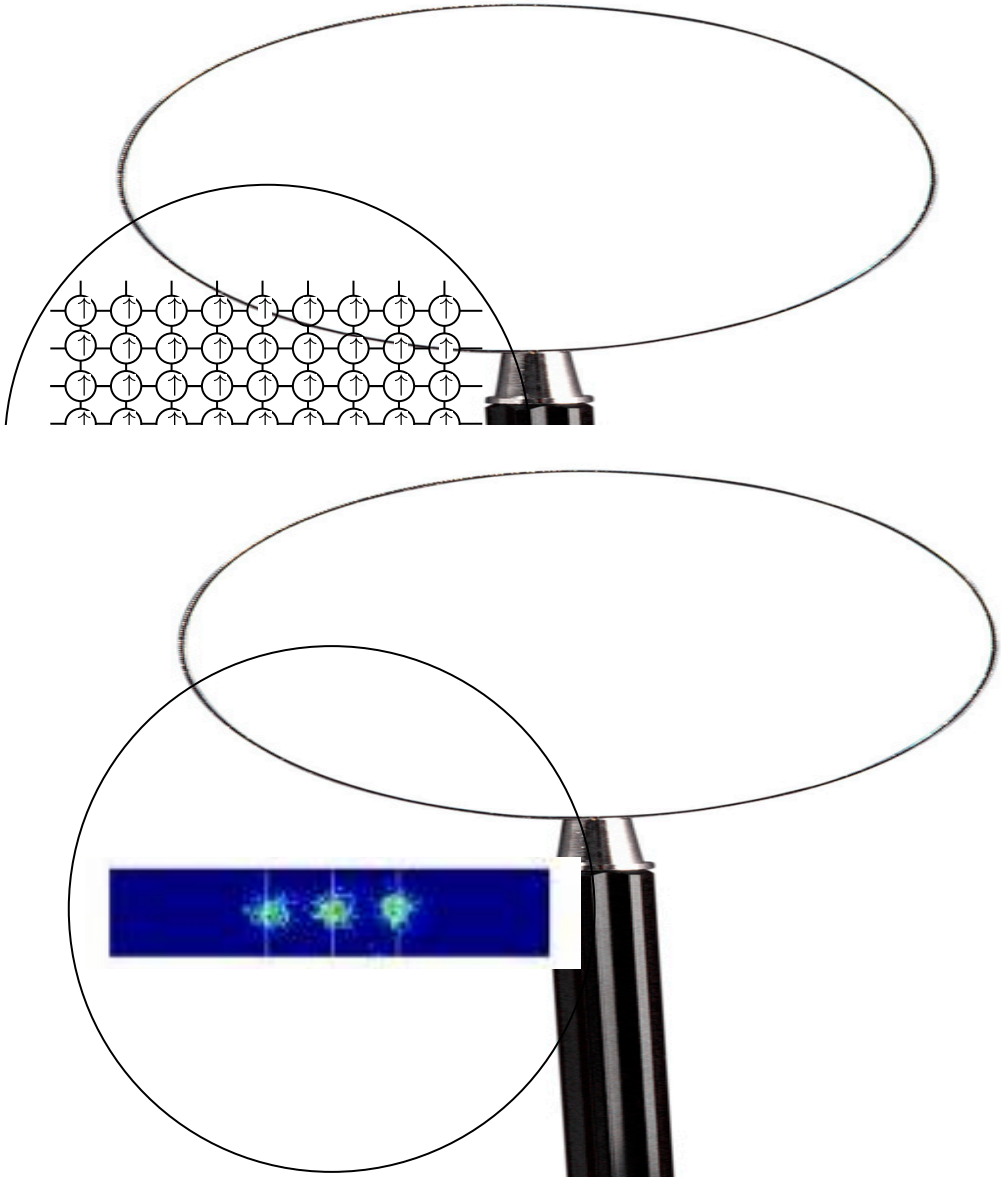
No Information without Physical Representation!







NIST Bo



Noisy classical

Robust classical

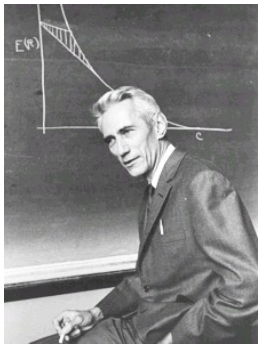
Noisy quantum

?

Robust quantum

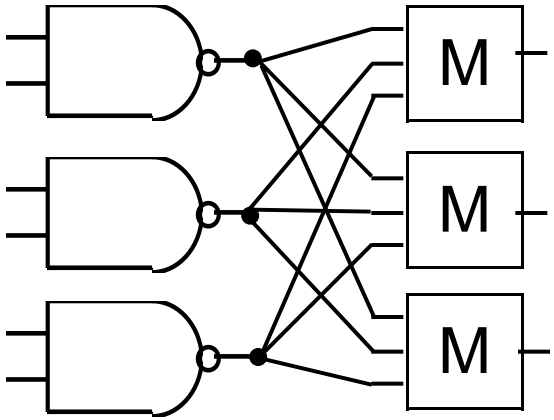
Classical Fault Tolerance….Theory

error correcting codes + fault tolerant construction



Redundancy to protect information against noise

Claude Shannon 1948
“A mathematical theory of communication”

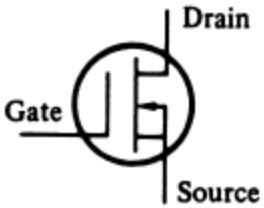




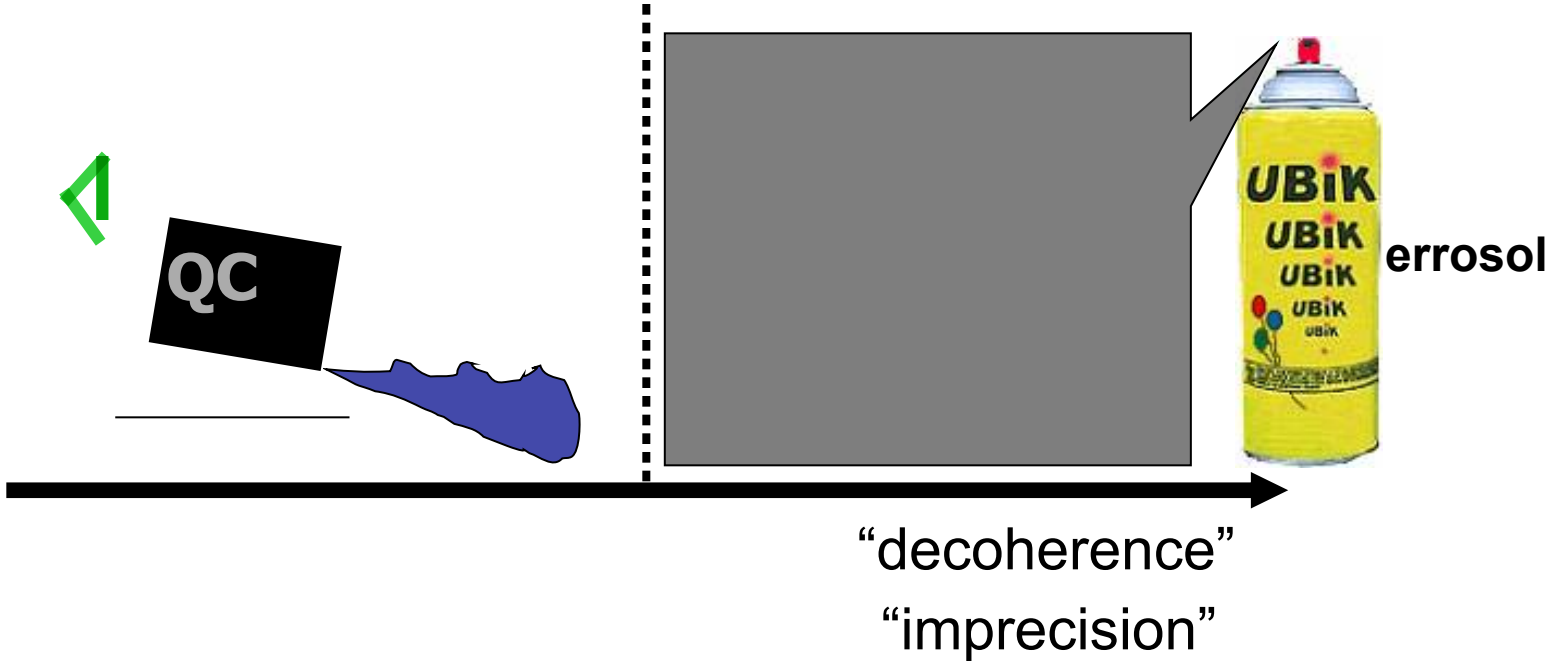
John von Neumann, 1956

“Probabilistic logics and the synthesis of reliable organisms from unreliable components”

Preempt:



Quantum Threshold Theorem

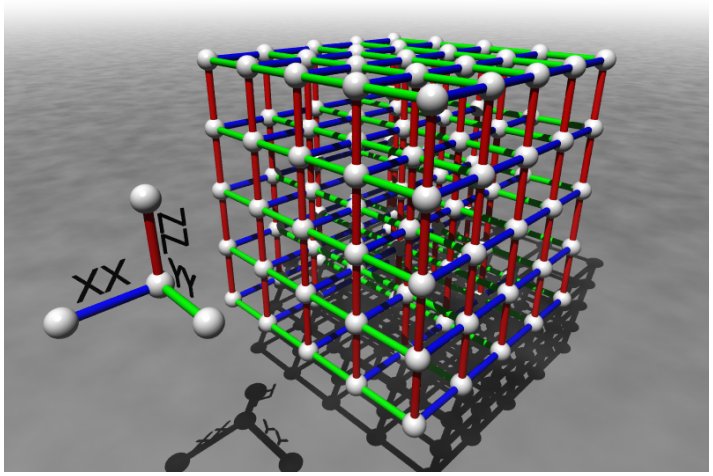


Threshold Theorem for Fault-Tolerant Quantum Computation

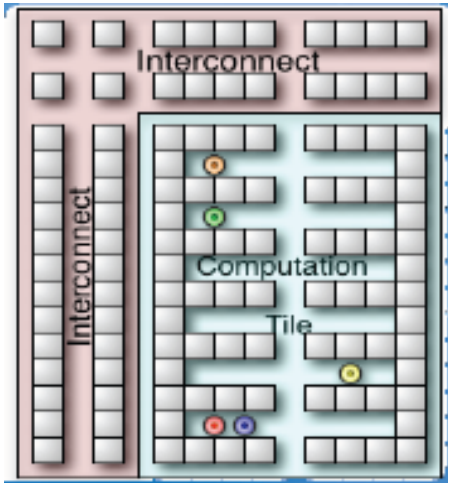
[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Kitaev, Gottesman, Preskill, Shor, + others]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Turning Theorems Into Devices



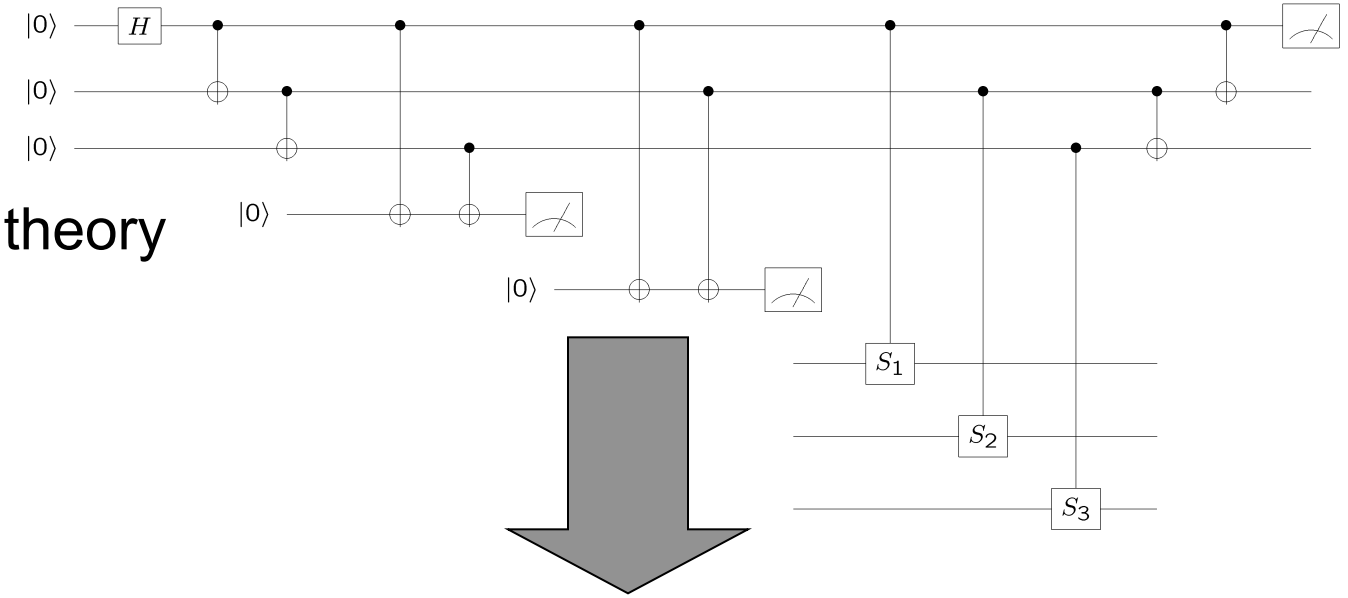
Natural fault-tolerant quantum computing



Quantum error correcting architectures

Quantum Architectures

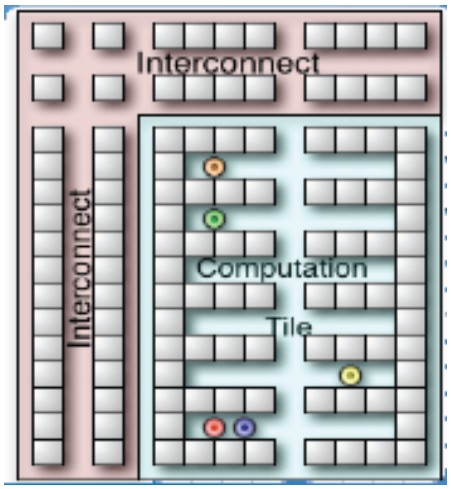
Next few years = scaling to 10-100s of qubits



Abstract theory

How to build, optimize

Tool chain for quantum architectures



Ion traps, Phosphorus donors in Silicon, Superconducting circuits,….

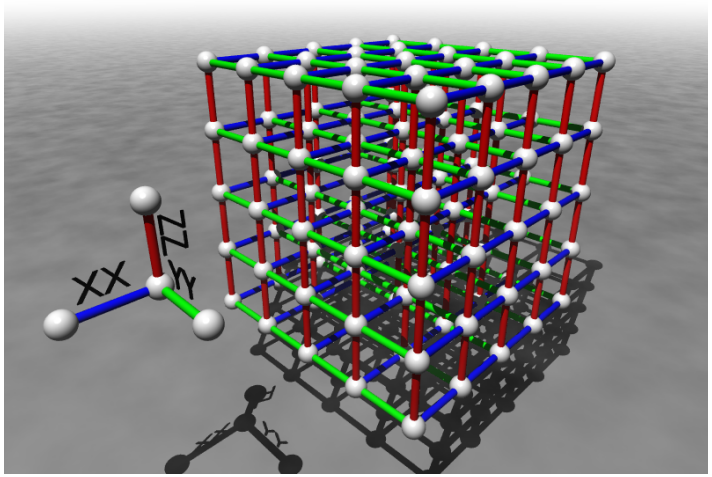
Naturally Fault-Tolerance

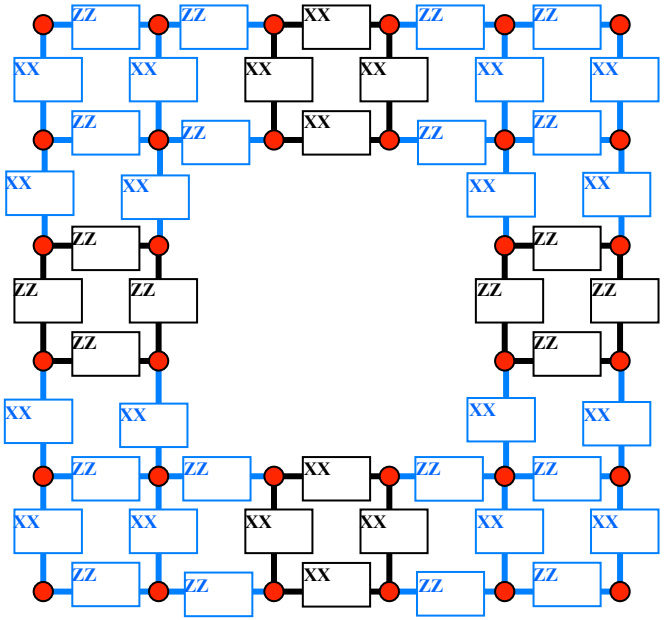


Physics of device responsible for robustness

Cooling = Error Correction

Duplicate for quantum devices?





Leapfrog to robust devices?

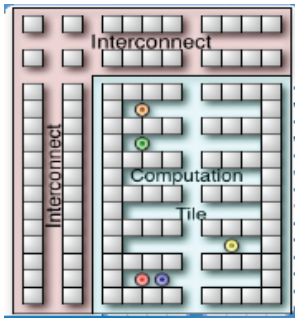
Quantum Computing at UW

QWiQ

group

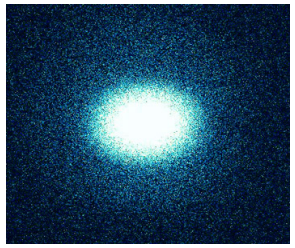
Dave Bacon (CSE)

Quantum Architectures



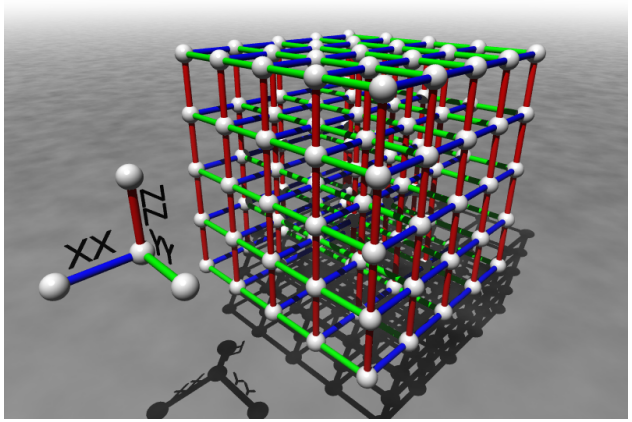
Mark Oskin (CSE)

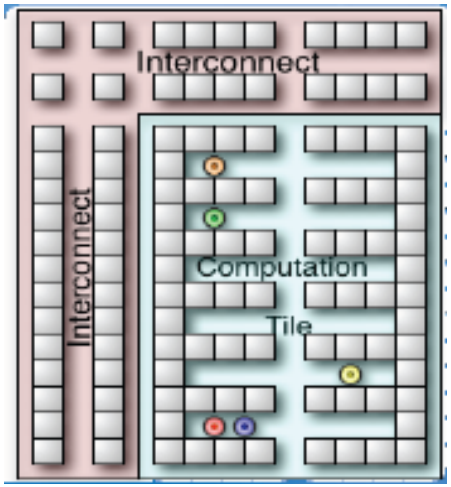
Ion Trap
Quantum Computers



Boris Blinov (Physics)

How to Build a Quantum Computer





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ARMY RESEARCH OFFICE
1951

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